ESP32 Motor Control Command Documentation

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Overview

This document details the commands for the `grok_2ppm_2servo_2ain_r1.ino` firmware, controlling an ESP32-based motor control system. The system manages two motors (via ESCs on GPIO 14 and 27), two servos (GPIO 12 and 13 for string release), two analog inputs (GPIO 36 and 39), and a built-in LED (GPIO 2). Commands are sent via serial (9600 baud) using 2, 3, or 4 characters, case-insensitive, terminated by Enter (`\r` or `\n`). The system initializes with neutral PWM (1500 µs) and a default PPM level of `mx 5` (forward: 1694 µs, reverse: 1306 µs). Each command flashes the LED and logs a response to serial. Every 10th command logs "Command log: Last 10 commands processed".

Command Summary

Command	Length	Description
`ff, fr, fh, rf, rr, rh, hf, hr, hh`	2 chars	Set motor PWM (GPIO 14/27); update servos from analog inputs.
`x1, x2`	2 chars	Release servo 1 (GPIO 12) or servo 2 (GPIO 13).
`ai`	2 chars	Read analog inputs (GPIO 36/39).
`ain`	3 chars	Read analog inputs (alternative).
`st`	2 chars	Stop all motors/servos (1500 µs).
`sl`	2 chars	Swap left/right motor outputs.
`sf`	2 chars	Swap forward/reverse motor outputs.
`mx <1..10>`	4 chars	Set PWM range (forward: 1450 µs to 2000 µs, reverse: 1550 µs to 1000 µs).

Table 1: Command Summary for `grok_2ppm_2servo_2ain_r1.ino`

Detailed Command Descriptions

Motor Control Commands (2 Characters: XY)

- **Function**: Sets PWM for left (GPIO 14) and right (GPIO 27) motors; updates servos from analog inputs (GPIO 36/39).
- **Format**: `XY`, where `X` (left motor) and `Y` (right motor) are:
  - `F`: Forward (`maxForwardPPM`, e.g., 1694 µs for `mx 5`).

  - `R`: Reverse (`minReversePPM`, e.g., 1306 µs for `mx 5`).
  - `H`: Halt (1500 µs).
- **Commands**:
  - `ff`: Both forward (left/right: `maxForwardPPM`).
  - `fr`: Left forward, right reverse.
  - `fh`: Left forward, right halt.
  - `rf`: Left reverse, right forward.
  - `rr`: Both reverse (left/right: `minReversePPM`).
  - `rh`: Left reverse, right halt.
  - `hf`: Left halt, right forward.
  - `hr`: Left halt, right reverse.
  - `hh`: Both halt (left/right: 1500 µs).
- **Behavior**: Updates `leftPulseWidth`, `rightPulseWidth`; writes PWM to LEDC channels (0 for GPIO 14, 1 for GPIO 27) using `map(pulseWidth, 1000, 2000, 0, 1023)`. Reads analog inputs (0–4095) and maps to servo PWM (1000 µs to 2000 µs) for GPIO 12/13.
- **Output**: Serial: `Cmd: XY → Left=<leftPulseWidth>µs, Right=<rightPulseWidth>µs, Servo1=<servo1PulseWidth>µs, Servo2=<servo2PulseWidth>µs`. E.g., `Cmd: ff → Left=1694µs, Right=1694µs, Servo1=1500µs, Servo2=1500µs`.
- **Errors**: Invalid X/Y: `Invalid command: XY (use 2 chars: ff, fr, fh, rf, rr, rh, hf, hr, hh, x1, x2, ai, st, sl, sf; 3 chars: ain; or 4 chars: mx <1..10>)`.

Servo Release Commands (2 Characters: x1, x2)

- **Function**: Moves servo 1 (GPIO 12) or servo 2 (GPIO 13) from neutral (1500 µs) to `OPEN_POSITION` (1000 µs) and back, with ∼2.5 s transitions.
- **Commands**:
  - `x1`: Releases servo 1.
  - `x2`: Releases servo 2.
- **Behavior**: Sweeps PWM (1500 µs to 1000 µs, 5 ms/step, ∼2.5 s), holds for 500 ms, sweeps back. Uses LEDC channels (2 for GPIO 12, 3 for GPIO 13).
- **Output**: Serial: `Cmd: x1 → Servo 1 released with slow movement` or `Cmd: x2 → Servo 2 released with slow movement`.
- **Errors**: Invalid second character: `Invalid command: xN (use x1 or x2)`.

Analog Read Commands (2 or 3 Characters: ai, ain)

- **Function**: Reads analog inputs from GPIO 36 (VP, ADC1_CH0) and GPIO 39 (VN, ADC1_CH3).
- **Commands**:
  - `ai`: Reads analog inputs.

  - `ain`: Reads analog inputs (alternative).
- **Behavior**: Returns raw ADC values (0–4095).
- **Output**: Serial: `Cmd: ai → Analog1 (VP, GPIO 36)=<value>, Analog2 (VN, GPIO 39)=<value>`. E.g., `Cmd: ai → Analog1 (VP, GPIO 36)=2048, Analog2 (VN, GPIO 39)=2048`.

Stop Command (2 Characters: st)

- **Function**: Sets all motors and servos to neutral (1500 µs).
- **Command**: `st`
- **Behavior**: Sets `leftPulseWidth`, `rightPulseWidth`, `servo1PulseWidth`, `servo2PulseWid` to 1500 µs; writes to LEDC channels (0, 1, 2, 3).
- **Output**: Serial: `Cmd: st → All motors and servos stopped (1500µs).`

Swap Commands (2 Characters: sl, sf)

- **Function**: Modifies motor PWM outputs (GPIO 14/27) if in valid states (1500 µs, `maxForwardPPM`, `minReversePPM`).
- **Commands**:
  - `sl`: Swaps left/right outputs (GPIO 14 gets `rightPulseWidth`, GPIO 27 gets `leftPulseWidth`
  - `sf`: Swaps forward/reverse (`maxForwardPPM` ↔ `minReversePPM`).
- **Behavior**: Uses LEDC channels (0, 1).
- **Output**:
  - `sl`: `Cmd: sl → Motor outputs swapped: Left pin=<rightPulseWidth>µs, Right pin=<leftPulseWidth>µs.`
  - `sf`: `Cmd: sf → Fwd/rev outputs swapped: Left pin=<leftOutput>µs, Right pin=<rightOutput>µs.`
- **Errors**: Invalid motor state: `Cmd: sl → Swap failed: Motors must be set by a valid command` (or similar for `sf`).

Max PPM Command (4 Characters: mx <1..10>)

- **Function**: Sets PWM range (forward: 1450 µs to 2000 µs, reverse: 1550 µs to 1000 µs); updates motors if in forward/reverse state.
- **Command**: `mx N` (N = 1 to 10, e.g., `mx 1`, `mx 10`).
- **Behavior**: Calculates `maxForwardPPM = 1450 + (550 * (level - 1)) / 9`, `minReverse = 1550 - (550 * (level - 1)) / 9`. Updates `leftPulseWidth`/`rightPulseWidth` if needed. Examples:
  - `mx 1`: Forward=1450 µs, Reverse=1550 µs.
  - `mx 5`: Forward=1694 µs, Reverse=1306 µs (default).
  - `mx 10`: Forward=2000 µs, Reverse=1000 µs.

- **Output**: Serial: `Cmd: mx N → Max PPM set: Forward=<maxForwardPPM>µs, Reverse=<minReversePPM>µs`.
- **Errors**: Invalid format/number: `Cmd: mx → Current max PPM: <maxForwardPPM>µs (level <maxPPMLevel>)`.

Notes

- **Serial**: Commands processed in `loop()`, accumulated in `inputString` until newline.
- **PWM**: 50 Hz, 10-bit resolution; maps 1000 µs to 2000 µs to 0–1023.
- **Hardware**: GPIO 14/27 (ESCs), 12/13 (servos), 36/39 (resistors for solar controller), 2 (LED).
- **Errors**: Invalid length/format: `Invalid command length: <input> (expect 2, 3, or 4 chars)`.